

New q -ary Sequential Locally Recoverable Codes from Product Codes

Akram Baghban¹, Marc Newman², Anna-Lena Horlemann³, and Mehdi Ghiyasvand⁴

^{1,4}Bu-Ali Sina University, Hamedan, Iran^{*}

^{2,3}University of St.Gallen[†]

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Abstract

This work focuses on sequential locally recoverable codes (SLRCs), a special family of locally repairable codes, capable of correcting multiple code symbol erasures, which are commonly used for distributed storage systems. First, we construct an extended q -ary family of non-binary SLRCs using product codes with a novel maximum number of recoverable erasures t and a minimal repair alternativity A . Second, we study how MDS and BCH codes can be used to construct q -ary SLRCs. Finally, we compare our codes to other locally recoverable codes.

Key Words: Distributed storage, Locally repairable codes, Erasure correction, Sequential recovery.

1 Introduction

In a distributed storage system (DSS) data is stored in a large, distributed network of nodes or storage units. To protect the system against node failure, current DSS systems utilize different coding schemes—specifically locally recoverable codes (LRCs)—which have recently been the subject of study. The purpose of an LRC is to reduce the total number of nodes required for node repair. Generally, an LRC with locality r is a linear code over a finite field where the value of every code symbol is recoverable by accessing at most r other symbols.

^{*}a.baghban@sci.basu.ac.ir, mghiyasvand@basu.ac.ir

[†]marc.newman@unisg.ch, anna-lena.horlemann@unisg.ch

In such systems, it is common for multiple storage nodes to fail simultaneously. To address this, two primary approaches have been developed for locally recoverable codes capable of handling multiple erasures: parallel recovery and sequential recovery. In the parallel approach, each erased code symbol must be recovered independently using only the symbols that remain uncorrupted. In contrast, the sequential approach allows erased symbols to be ordered and recovered one at a time with symbols being reconstructed using both the initially available symbols and those subsequently recovered.

This sequential strategy offers greater flexibility than its parallel counterpart, as the choice of recovery sets can adapt dynamically throughout the process and therefore one can potentially handle a larger number of erasures. Nevertheless, codes that support parallel recovery are often favored in practice due to their ability to reconstruct multiple erased symbols simultaneously, thereby reducing recovery latency and enhancing performance in scenarios with high read demand. The interested reader is referred to [GHSY12, HCL13, KPLK14, PD14, TBF16] for more information on locally recoverable codes and [BPK16, PLBK19, RMV15, SCY⁺18] for sequential LRCs.

This research introduces a q -ary family of non-binary sequential locally recoverable codes (SLRCs) which are capable of recovering $t \geq 2$ erasures having small locality r . With our structure we establish upper bounds on the total number of recoverable erasures and code rate which improve the results from [BPK16], [SCY⁺18], and [WZL15] using product codes. Additionally, we achieve a broader range of possible code parameters compared to other known results.

The paper is structured as follows: Section 2 presents necessary preliminaries, in Section 3 we provide our general q -ary construction for SLRCs, in Section 4 we give explicit constructions using products of MDS and BCH codes, and in Section 5 we compare our constructions to other known constructions of general LRCs. The proofs of the lemmas and theorems in Section 3 can be found in the extended version of this work [BNHG25].

2 Preliminaries

We fix some notation that will be used throughout. For any subset A , $|A|$ is the cardinality of A . A set B is referred to as a t -subset of A if $B \subseteq A$ and $|B| = t$. For any positive integer n , we denote by $[n]$ the set $\{1, 2, \dots, n\}$. The set $\mathbb{F}_q$ is the finite field with q elements. Furthermore, let $\text{supp}(x)$ denote the support of a vector x .

A q -ary **linear code** of length n and dimension k , denoted an $[n, k]_q$ code, is a dimension k linear subspace of the vector space $\mathbb{F}_q^n$ whose **rate** is defined to be $\frac{k}{n}$. The elements of a code are referred to as its codewords. The **dual** of a linear code $\mathcal{C}$, denoted $\mathcal{C}^\perp$, consists of all vectors orthogonal

to all codewords of $\mathcal{C}$; it is itself a linear code of dimension $n - k$. The **Hamming weight** $\text{wt}(x)$ of a vector x is defined as the number of non-zero entries of x , i.e., $\text{wt}(x) = |\text{supp}(x)|$. The **Hamming distance** $d(x, y)$ between two vectors x and y is the Hamming weight of their difference, i.e., $d(x, y) = \text{wt}(x - y)$. The **minimum distance** of a code $\mathcal{C}$, denoted by $d(\mathcal{C})$, is defined as the minimal distance between any two distinct codewords in the code, i.e., $d(\mathcal{C}) = \min\{d(x, y) \mid x, y \in \mathcal{C}, x \neq y\}$. In a linear code, this is equivalent to the minimal Hamming weight of any nonzero codeword in $\mathcal{C}$. If an $[n, k]_q$ code has minimum distance d , we denote it as an $[n, k, d]_q$ code.

Definition 1. [SCY⁺18] Let $i \in [n]$ and $R_i \subseteq [n] \setminus \{i\}$. The subset R_i is called an $(r, \mathcal{C})$ -**recovery set** of i if $|R_i| \leq r$ and there exist $a_{ij} \in \mathbb{F}_q$ such that $x_i = \sum_{j \in R_i} a_{ij} x_j$ for all $(x_1, \dots, x_n) \in \mathcal{C}$.

We say a code $\mathcal{C}$ has **locality** r if every symbol of $\mathcal{C}$ has a $(r, \mathcal{C})$ -recovery set. An $[n, k, d]_q$ code with locality r is referred to as an (n, k, r) -**locally recoverable code (LRC)**.

Many constructions of LRCs exist and bounds on the rate and a Singleton-like bound have been given for them, see, e.g., [TB14]. An LRC with equality in the Singleton-like bound is referred to as an **optimal LRC code**. This optimality is with respect to the minimum distance of a locally recoverable code. If we have disjoint recovery sets, then the information theoretic optimal codes are **partial maximum distance separable (PMDS) codes**. In other words, a code is PMDS if the projection to any recovery set is an MDS code, and if additionally we can recover the maximal amount of global erasures, i.e., additional erasures anywhere in the code, see, e.g., [HTN20].

In this paper we are mainly interested in the case where we do not have disjoint recovery sets and where sequential recovery is beneficial over parallel recovery. We explain these concepts in the following.

Definition 2. Let E be a t -subset of $[n]$. $\mathcal{C}$ is said to be (E, r) -**recoverable** if there exists a sequential indexing of the elements of E , say $(i_1, i_2, \dots, i_t)$, such that each $i_j \in E$ has an $(r, \mathcal{C})$ -recovery set $R_{i_j} \subseteq \overline{E} \cup \{i_1, i_2, \dots, i_{j-1}\}$ of size $|R_{i_j}| \leq r$ where $\overline{E} = [n] \setminus E$. An $[n, k, d]_q$ -linear code $\mathcal{C}$ is said to be an (n, k, r, t) -**sequential locally recoverable code (SLRC)** if for each $E \subseteq [n]$ of size $|E| \leq t$, $\mathcal{C}$ is (E, r) -recoverable.

Note that t is the upper bound of the number of (sequential locally) recoverable failed nodes and that by definition we have $2 \leq |R_i| \leq r \leq k < n$ where R_i is a recovery set for some (n, k, r, t) -SLRC. Additionally, any (n, k, r, t) -SLRC can also be considered as an (n, k, r', t) -SLRC for $r < r' \leq k$.

Proposition 3. [SCY⁺18] $\mathcal{C}$ is an (n, k, r, t) -SLRC if and only if for any nonempty $E \subseteq [n]$ of size $|E| \leq t$ there exists an $i \in E$ such that i has an $(r, \mathcal{C})$ -recovery set $R_i \subseteq [n] \setminus E$.

Another interesting parameter of locally recoverable codes is its repair alternativity, which refers to the number of different recovery sets available for recovering lost data:

Definition 4. [PJHO13] Let $\mathcal{C}$ be an (n, k, r, t) -SLRC and for $i \in [n]$ let $\Omega_r(i) = \{v \in \mathcal{C}^\perp \mid i \in \text{supp}(v), \text{wt}(v) \leq r + 1\}$. We say the **repair alternativity** of $i \in [n]$ is $a(i) = |\{\text{supp}(v) \mid v \in \Omega_r(i)\}|$ and that the **repair alternativity** of the code $\mathcal{C}$ is $a = \min_{i \in [n]} \{a(i)\}$. We call the elements of $\Omega_r(i)$ the **recovery vectors** for coordinate i in the code $\mathcal{C}$.

It is easy to see and well-known that an $[n, k, d]_q$ -MDS code has locality k and alternativity $\binom{n-1}{k}$.

Remark 5. A concept similar to alternativity is **availability**, which is often used in the context of parallel recovery. However, in the setting of sequential local recovery, alternativity is more relevant. While availability quantifies the number of disjoint recovery sets for simultaneous repairs in parallel recovery, sequential recovery relies on the adaptive use of previously recovered symbols, making alternativity the appropriate measure for characterizing the flexibility in the recovery process.

As our code construction mainly relies on the Kronecker product of generator matrices of codes, we will define it and state its main properties in what follows.

Definition 6. Let $A = (a_{ij})_{i,j}$ be an $m_1 \times n_1$ matrix and let B be an $m_2 \times n_2$ matrix. Then the **Kronecker product** $A \otimes B$ is the $m_1 m_2 \times n_1 n_2$ matrix

$$A \otimes B = \begin{bmatrix} a_{11}B & \cdots & a_{1n_1}B \\ \vdots & \ddots & \vdots \\ a_{m_1 1}B & \cdots & a_{m_1 n_1}B \end{bmatrix}.$$

Proposition 7. [HJ94] Let A, B, C , and D be matrices such that AC and BD are valid products. Then

- $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$,
- $(A \otimes B)^\top = A^\top \otimes B^\top$.

Lemma 8. Let $\mathbb{K}$ be an arbitrary field and let $u \in \mathbb{K}^m$ and $v \in \mathbb{K}^n$ be vectors. Then if $i \in [mn]$, there exists a unique pair $(j, k) \in [m] \times [n]$ such that $i = (j - 1)n + k$ and $i \in \text{supp}(u \otimes v)$ if and only if $j \in \text{supp}(u)$ and $k \in \text{supp}(v)$.

Definition 9. [MS77] Let $\mathcal{A}$ and $\mathcal{B}$ be $[n_1, k_1]_q$ and $[n_2, k_2]_q$ linear codes with generator matrices G_A and G_B , respectively. Then the **product code** $\mathcal{A} \otimes \mathcal{B}$ is the $[n_1 n_2, k_1 k_2]_q$ linear code with generator matrix $G_A \otimes G_B$. Furthermore, the product code of codes with minimum distances d_1 and d_2 has minimum distance $d_1 d_2$.

We remark that the codewords of $\mathcal{A} \otimes \mathcal{B}$ can be represented as $n_1 \times n_2$ matrices such that each row forms an element of $\mathcal{B}$ and each column is from $\mathcal{A}$.

3 General q -ary SLRC Construction

We will now present and analyze our construction for sequential locally recoverable codes from the code product. We will first state it in all generality before constructing explicit codes with this machinery in the following section.

Our code construction is the ℓ -fold product code of locally recoverable codes with the same locality parameter r and is a generalization of a similar construction in [BPK16], where the authors restrict to the case of $q = \ell = 2$. The lengths, dimensions and erasure correction capabilities of the small codes can be variable. Depending on the small codes' parameters we can determine the corresponding parameters of the product code, as we show in Theorem 11 and Corollary 12. To prove the main result we need the following lemma:

Lemma 10. *For $i \in \{1, 2\}$, let $\mathcal{C}_i$ be an (n_i, k_i, r_i, t_i) -SLRC over a field $\mathbb{F}_q$, and let $e_m^{(i)}$ be the m th unit vector in $\mathbb{F}_q^{n_i}$.*

- (a) *For any recovery vector $c^{(1)} \in \mathcal{C}_1^\perp$ for coordinate $j \in [n_1]$ in the code $\mathcal{C}_1$ of weight $\bar{r} \leq r_1 + 1$, the vector $c^{(1)} \otimes e_m^{(2)}$ is a recovery vector for coordinate $(j - 1)n_2 + m$ in the code $\mathcal{C}_1 \otimes \mathcal{C}_2$ of weight $\bar{r}$.*
- (b) *For any recovery vector $c^{(2)} \in \mathcal{C}_2^\perp$ for coordinate $j \in [n_2]$ in the code $\mathcal{C}_2$ of weight $\bar{r} \leq r_2 + 1$, the vector $e_m^{(1)} \otimes c^{(2)}$ is a recovery vector for coordinate $(m - 1)n_2 + j$ in the code $\mathcal{C}_1 \otimes \mathcal{C}_2$ of weight $\bar{r}$.*

Theorem 11. *For $i \in \{1, 2\}$, let $\mathcal{C}_i$ be an (n_i, k_i, r_i, t_i) -SLRC over a field $\mathbb{F}_q$ with repair alternativity a_i and let $N = n_1 n_2, K = k_1 k_2, T = t_1 t_2 + t_1 + t_2, r = \max\{r_1, r_2\}$ and $A = a_1 + a_2$. Then $\mathcal{C}_1 \otimes \mathcal{C}_2$ is a q -ary (N, K, r, T) -SLRC with repair alternativity $\geq A$. Moreover, if $t_i + 1$ is equal to the minimum distance of $\mathcal{C}_i$ (for $i = 1, 2$), then $T + 1$ is equal to the minimum distance of $\mathcal{C}_1 \otimes \mathcal{C}_2$. This means that the local recovery capacity stays maximal under the code product.*

We note that the above result applies only to the case of sequential local recovery, not parallel local recovery. This is because the argument concerning the erasure distribution ensures only that at least one block corresponding to $\mathcal{C}_2$ contains at most t_2 errors. In contrast, parallel recovery requires that all such blocks contain at most t_2 errors. An analogous issue arises for (spread out) blocks associated with $\mathcal{C}_1$, which would each need to contain no more than t_1 errors.

The following follows by applying Theorem 11 iteratively.

Corollary 12. For all $i \in [\ell]$, let $\mathcal{C}_i$ be an (n_i, k_i, r_i, t_i) -SLRC over $\mathbb{F}_q$ with repair alternativity a_i and let $r_\ell = \max_{i \in [\ell]} \{r_i\}$,

$$N_\ell = \prod_{i=1}^{\ell} n_i, \quad K_\ell = \prod_{i=1}^{\ell} k_i, \quad T_\ell = \left(\prod_{i=1}^{\ell} (t_i + 1) \right) - 1,$$

and $A_\ell = \sum_{i=1}^{\ell} a_i$. Then $\mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_\ell$ is a q -ary $(N_\ell, K_\ell, r_\ell, T_\ell)$ -SLRC with repair alternativity $A \geq A_\ell$ and rate $\frac{K_\ell}{N_\ell}$. Moreover, if $t_i + 1 = d(\mathcal{C}_i)$ for all $i \in [\ell]$, then $T_\ell + 1 = d(\mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_\ell)$.

For a geometric approach of the code product for sequential recovery, we refer the interested reader to the extended version of this work [BNHG25]. Furthermore, we show that the lower bound on the alternativity is indeed met for some code families.

Remark 13. The construction of recovery vectors for the product code from those of the component codes naturally leads to an algorithmic procedure for erasure recovery. Given an erasure pattern, one identifies projections onto component codes such that the erased coordinates on these projections can be recovered locally via the associated recovery vectors. This process is applied iteratively, updating the erasure pattern after each successful recovery step, until all erasures are resolved.

For correcting erasures, consider an $(n_1 \cdots n_\ell, k_1 \cdots k_\ell, r, t)$ -SLRC $\mathcal{C} = \mathcal{C}_1 \otimes \cdots \otimes \mathcal{C}_\ell$ where $\mathcal{C}_j$ is an $[n_j, k_j]_q$ code for each $1 \leq j \leq \ell$. We index the symbols of $\mathcal{C}$ by $(i_1, \dots, i_\ell)$ where each $i_j \in [n_j]$. Then, given some erased symbol x at $i_1, i_2, \dots, i_\ell$, for each $1 \leq j \leq \ell$ we can recover x using any r -recovery set $R \subset \mathcal{C}_j$ where $i_j \in \text{supp}(R)$.

Theorem 14. Let $\mathcal{C}$ be an $[n, k, d]_q$ code with $1 \leq k < n$, let $\ell \geq 2$, and let μ denote the number of erasures. Then the ℓ -fold product $\mathcal{C}^* := \mathcal{C} \otimes \cdots \otimes \mathcal{C}$ has the following erasure recovery behavior:

- (a) If $\mu \leq \ell(d - 1)$, we can recover everything with only one copy of the $[n, k, d]_q$ code (in parallel).
- (b) If $\ell(d - 1) < \mu < d^\ell$, we can fully sequentially recover, but we cannot necessarily recover in parallel.
- (c) If $d^\ell \leq \mu \leq \min(n^\ell - k^\ell, \ell n^{\ell-1}(d - 1))$, we can possibly recover depending on the erasure pattern.
- (d) If $\min(n^\ell - k^\ell, \ell n^{\ell-1}(d - 1)) < \mu$, we cannot fully recover.

4 Product Codes from MDS and BCH Codes

In this section, we derive several examples to demonstrate the achievable parameters of our construction by using MDS and BCH codes as the building blocks for the product code.

For this let $\mathcal{P}_q$ be a $[q+1, 2, q]_q$ MDS code, $\mathcal{D}_{q,n}$ be a $[n, n-1, 2]_q$ MDS code, and $\mathcal{R}_{q,n,k}$ be a $[n, k, n-k+1]_q$ Reed-Solomon code. Furthermore, denote by $\mathcal{B}_{q,n,d}$ a BCH code over $\mathbb{F}_q$ with length n and designed distance d . For some code $\mathcal{C}$, let $\mathcal{C}^{*S}$ denote the code $\mathcal{C}$ punctured at indices S .

The achievable parameters are presented in the tables below, where we omit certain product codes when a superior code exists for the same values of q and k . By “superior”, we refer to codes that exhibit one or more of the following characteristics: higher ℓ , lower length n , lower locality r , or a higher number of recoverable erasures t . For instance, in $q = 5$, we omit the $(16, 6, 3, 5)$ -SLRC because the $(15, 6, 3, 5)$ -SLRC achieves the same parameters with a smaller length. Additionally, we omit constructions from codes whose dual codes have minimum distance 2 since in most cases given the parameters of the resulting product codes, optimal codes are known to exist [GXY19].

Tables 1 and 2 give parameter sets for products of MDS and (possibly punctured) BCH codes over $\mathbb{F}_3$ and $\mathbb{F}_5$ up to $k = 10$. For more achievable parameters and graphic illustrations of them we refer again to [BNHG25].

5 Comparison to other LRCs

In the general literature on LRCs, most work focuses on either parallel recovery, binary alphabets, or both. Additionally, much of the work focuses on codes with a small number of erasures (e.g., $t = 2$). In contrast, our construction is very general and works for any field size q and number of erasures t . We note that optimal constructions for LRCs exist, in particular for large n relative to r and d (for example, see [GXY19]). In particular, without loss of generality, whenever

$$\frac{n}{r+1} \geq \left(d - 2 - \left\lfloor \frac{d-2}{r+1} \right\rfloor \right) (3r+2) + \left\lfloor \frac{d-2}{r+1} \right\rfloor + 1$$

one can assume that optimal LRCs have disjoint recovery sets and can then recover all recoverable erasure patterns with parallel recovery [GXY19]. In the case of the parameters using our construction, this bound is never met, meaning that sequential recovery can be better than parallel recovery.

Additionally, Tamo and Barg [TB14] gave optimal constructions using evaluations of polynomials for the case where $q \geq n$ and when there exists a “good” polynomial for the desired value of r . The results were later extended to codes with high availability—an analog of alternativity in the parallel recovery setting—in [GMN⁺23] for some parameter sets using codes from

Table 1. Parameters for $q = 3, k < 10, n < 50$

q	k	r	ℓ	Code	t	$a \geq$	Construction
3	4	2	2	[16, 4, 9]	8	6	$\mathcal{P}_3 \otimes \mathcal{P}_3$
				[12, 4, 6]	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,3}$
				[9, 4, 4]	3	2	$\mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$
3	6	2	2	[32, 6, 15]	14	6	$\mathcal{P}_3 \otimes \mathcal{B}_{3,8,5}$
				[24, 6, 10]	9	4	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,5}$
3	6	3	2	[16, 6, 6]	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,4}$
				[12, 6, 4]	3	2	$\mathcal{D}_{3,4} \otimes \mathcal{D}_{3,3}$
3	8	4	2	[20, 8, 6]	5	4	$\mathcal{P}_3 \otimes \mathcal{D}_{3,5}$
				[15, 8, 4]	3	2	$\mathcal{D}_{3,5} \otimes \mathcal{D}_{3,3}$
3	8	3	2	[32, 8, 12]	11	8	$\mathcal{P}_3 \otimes \mathcal{B}_{3,8,4}$
				[28, 8, 9]	8	5	$\mathcal{P}_3 \otimes \mathcal{B}_{3,8,4}^{*\{1\}}$
				[24, 8, 8]	7	6	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,4}$
				[21, 8, 6]	5	3	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,4}^{*\{1\}}$
				[18, 8, 4]	3	3	$\mathcal{D}_{3,3} \otimes \mathcal{B}_{3,8,4}^{*\{1,5\}}$
3	8	2	3	[48, 8, 18]	17	7	$\mathcal{P}_3 \otimes \mathcal{P}_3 \otimes \mathcal{D}_{3,3}$
				[36, 8, 12]	11	5	$\mathcal{P}_3 \otimes \mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$
				[27, 8, 8]	7	3	$\mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3} \otimes \mathcal{D}_{3,3}$
3	9	3	2	[32, 9, 10]	9	4	$\mathcal{D}_{3,4} \otimes \mathcal{B}_{3,8,5}$
				[16, 9, 4]	3	2	$\mathcal{D}_{3,4} \otimes \mathcal{D}_{3,4}$

evaluations of polynomials on paradoxical families of subsets. Therefore, we focused on constructing codes filling in the (many) gaps outside of these categories, i.e., q -ary (for $2 < q < n$), sequentially recoverable codes with large d (or equivalently t) and small n relative to r .

In [PLBK19], (in their words, “rate-” and “distance-”) optimal SLRCs are constructed from both Turán and regular graphs. These specifically target the $t = 2$ case and gives codes with rate $\frac{r}{r+2}$. Similarly, using Turán hypergraphs, [BPK16] extended these results to the $t = 3$ case with rate $\frac{r}{r+3}$. In [BKK19], optimal binary SLRCs are built from Moore graphs—regular graphs whose girth is more than twice its diameter. However, these corresponding graphs only exist for $t \in \{2, 3, 4, 5, 7, 11\}$ when $r \geq 2$. Compared to them, our construction is more general and works for any q and t .

In [SY16], products of $[n, n - 1, 2]_2$ codes are used; our method generalizes this to other codes over q -ary fields. Following this, the authors of [SCY⁺18] derived a new family of specific instances of binary product codes which

Table 2. Parameters for $q = 5, k < 10, n < 50$

q	k	r	ℓ	Code	t	$a \geq$	Construction
5	4	2	2	[36, 4, 25]	24	20	$\mathcal{P}_5 \otimes \mathcal{P}_5$
				[30, 4, 20]	19	16	$\mathcal{P}_5 \otimes \mathcal{R}_{5,5,2}$
				[25, 4, 16]	15	12	$\mathcal{R}_{5,5,2} \otimes \mathcal{R}_{5,5,2}$
				[24, 4, 15]	14	13	$\mathcal{P}_5 \otimes \mathcal{R}_{5,4,2}$
				[20, 4, 12]	11	9	$\mathcal{R}_{5,5,2} \otimes \mathcal{R}_{5,4,2}$
				[18, 4, 10]	9	11	$\mathcal{P}_5 \otimes \mathcal{D}_{5,3}$
				[16, 4, 9]	8	6	$\mathcal{R}_{5,4,2} \otimes \mathcal{R}_{5,4,2}$
				[15, 4, 8]	7	7	$\mathcal{R}_{5,5,2} \otimes \mathcal{D}_{5,3}$
				[12, 4, 6]	5	4	$\mathcal{R}_{5,4,2} \otimes \mathcal{D}_{5,3}$
				[9, 4, 4]	3	2	$\mathcal{D}_{5,3} \otimes \mathcal{D}_{5,3}$
5	6	3	2	[30, 6, 15]	14	14	$\mathcal{P}_5 \otimes \mathcal{R}_{5,5,3}$
				[25, 6, 12]	11	10	$\mathcal{R}_{5,5,3} \otimes \mathcal{R}_{5,5,2}$
				[24, 6, 10]	9	11	$\mathcal{P}_5 \otimes \mathcal{D}_{5,4}$
				[20, 6, 9]	8	7	$\mathcal{R}_{5,5,3} \otimes \mathcal{R}_{5,4,2}$
				[15, 6, 6]	5	5	$\mathcal{R}_{5,5,3} \otimes \mathcal{D}_{5,3}$
				[12, 6, 4]	3	2	$\mathcal{D}_{5,4} \otimes \mathcal{D}_{5,3}$
5	8	4	2	[30, 8, 10]	9	11	$\mathcal{P}_5 \otimes \mathcal{D}_{5,5}$
				[25, 8, 8]	7	7	$\mathcal{R}_{5,5,2} \otimes \mathcal{D}_{5,5}$
				[20, 8, 6]	5	4	$\mathcal{R}_{5,4,2} \otimes \mathcal{D}_{5,5}$
				[15, 8, 4]	3	2	$\mathcal{D}_{5,5} \otimes \mathcal{D}_{5,3}$
5	8	2	3	[48, 8, 18]	17	7	$\mathcal{R}_{5,4,2} \otimes \mathcal{R}_{5,4,2} \otimes \mathcal{D}_{5,3}$
				[45, 8, 16]	15	8	$\mathcal{R}_{5,5,2} \otimes \mathcal{D}_{5,3} \otimes \mathcal{D}_{5,3}$
				[36, 8, 12]	11	5	$\mathcal{R}_{5,4,2} \otimes \mathcal{D}_{5,3} \otimes \mathcal{D}_{5,3}$
				[27, 8, 8]	7	3	$\mathcal{D}_{5,3} \otimes \mathcal{D}_{5,3} \otimes \mathcal{D}_{5,3}$
5	9	3	2	[25, 9, 9]	8	8	$\mathcal{R}_{5,5,3} \otimes \mathcal{R}_{5,5,3}$
				[20, 9, 6]	5	5	$\mathcal{R}_{5,5,3} \otimes \mathcal{D}_{5,4}$
				[16, 9, 4]	3	2	$\mathcal{D}_{5,4} \otimes \mathcal{D}_{5,4}$

give a rate of $(\sum_{s=0}^t r^{-|\text{supp}_\ell(s)|})^{-1}$, where $r \geq 2$, $t \geq 1$, ℓ is any positive integer satisfying $t \leq 2^\ell - 1$, and $\text{supp}_\ell(s)$ is the support of the ℓ -digit binary representation of s . However, this construction works only for $k = r^m$ and is hence more restricted than ours. The second binary construction of [SCY⁺18] achieves a rate of $r/(2r + t - 1)$ and has again restrictions on the parameters. In the following table we illustrate achievable rates of our construction compared to the binary constructions of [WZL15] and [SCY⁺18] for codes of locality $r = 2$, indicating the best rates in bold. The rates of our construction are taken from the examples in Section 4, for either $q = 3$ or

$q = 5$. We omit certain values of t (specifically ones where $t + 1$ is prime) since these cannot be directly constructed using products. Our rates are comparable to those of the first construction of [SCY⁺18] and hence beat the other two. Additionally, for fixed t , the rates of [SCY⁺18] are generally achieved with longer codes, e.g., for $r = 2$ they achieve $t = 5$ with a code of length 24 while ours achieves the same rate and erasure correction with $n = 12$.

Table 3. Achievable rates for locality $r = 2$.

rate $\setminus t$	3	5	7	8	9	11
this work	0.44	0.33	0.30	0.25	0.25	0.22
$\frac{r}{r+t}$ [WZL15]	0.40	0.29	0.22	0.20	0.18	0.15
1st const. [SCY ⁺ 18]	0.44	0.33	0.30	0.26	0.24	0.22
$\frac{r}{2r+t-1}$ [SCY ⁺ 18]	0.33	0.25	0.20	0.18	0.17	0.14

Finally, while Partial MDS (PMDS) codes are information-theoretically optimal LRCs, they can only locally recover up to t_i erasures in block i ; the remaining $s = mr - k$ erasures require global recovery using a recovery set of size k . In contrast, our construction provides locality for all erasure patterns. Moreover, PMDS codes typically require a field whose size is exponential in the number s of global erasures resulting in much larger fields than those used in this work.

Overall, the product code construction offers clear advantages in simplicity, generality of parameters, and improved rates compared to most existing constructions and, furthermore, we present clear lower bounds on the availability of our codes.

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